

Final Project Guide

Data Science for Healthcare

9,105 patients · UCI ML Repository ID 880 · 40+ variables

This is phase-by-phase roadmap for your project. Each phase contains some suggestions about what to do, what to think about, and what to document.

General suggestion

At the end of the project, you will have to write a short report and a poster to explain what you did. Documenting what you do throughout the whole project, at each step, will save you time at the end and, most importantly, give you a constant complete understanding of the problem you are solving.

Phase 0 Setup and Dataset Loading

The starter cells in the notebook handle this phase. Run them and verify the dataset loads correctly. Once you load the dataset via **ucimlrepo** you can inspect the X features and y targets. Using the provided cells, you can check the given metadata and a summary of all the variables. Here is where you may start to hypothesize some clinical questions.

Phase 1 Exploration and Dataset Understanding

Before running any analysis, it is necessary to properly understand the detailed content of the dataset. Here you must shape your knowledge about what types of variables you have, what the different variables mean, which can be useful for your analysis and how/why. There may be variables for which you are not able to understand the meaning: do not hesitate to ask, the '*clinical expert*' is there for you.

Suggestion: Before asking any question, try to bring a hypothesis first, like "We think this variable has this sense, is that correct?"

Phase 2 Formulating the Clinical Question

Good clinical questions follows these structure's examples:

"In patients with **[characteristics]**, is it possible to predict/identify/estimate **[outcome]**, using **[available information X]**?"

"Which **[characteristics]** do patients with **[certain outcome/ characteristic]** share?"

Your question determines: the target variable, which predictors are valid, the type of analysis (classification / regression / survival / clustering), and the right evaluation metric.

In this phase, to understand whether your question is interesting for clinical/research purposes, it can be useful to share your ideas with the *clinical expert*.

Phase 3 Data Cleaning and Preprocessing

Now you can prepare the dataset for analysis. Every cleaning/preprocessing decision has clinical implications, not just technical ones. Here you must check for missing values, understand how to deal with them, choose whether to reduce dimensionality etc etc.

Phase 4 Exploratory Data Analysis (EDA)

Now, explore the dataset in relation to your clinical question. EDA is where you develop the intuitions that will drive your modelling choices. There is no mandatory structure. You can compare the distribution of the chosen target among different categories, search for some particular predictors through correlation, understand how different variables may say the same thing, compare different patients groups.

In this phase, it can happen that you want to change your clinical question.

Phase 5 Feature Selection

At this point, you should have a pretty solid idea of what you want to look for through your analysis. It's time to build your statistical model, so you must decide how to use the variables you still have in the dataset. Is it necessary to keep all of them? Is it necessary to exclude some of them? Why?

*Always keep in mind the problem of **data leakage**.*

Phase 6 Modelling

Build and evaluate the model(s) appropriate to your clinical question. Choose evaluation metrics based on the clinical question. In medicine, false negatives and false positives often have very different costs. It may be not obvious what model to use, so you may want to compare some of them.

Phase 7 Explainability and Interpretation

A model in healthcare is more powerful if it can be interpreted. Analyse why the model makes the predictions it does. Are there any variables which are more important than others? Why? Are any variables characteristic of some patients subgroups?

Here, the clinical expert will be happy to hear your suppositions and give you his personal interpretation.

Phase 8 Conclusions

Step back and ask whether your work actually answers the question you set out to investigate. Is there anything you should have done differently? Can you still try to change your analyses or you just want to inform the expert about other possible strategies?

In this phase, it's time to wrap up also the limitations of the study. **Be honest**, don't care about impressing the clinical expert. Some well posed limitations open the path to further interesting analyses.

What have you understood during your analysis? Is it compatible with the literature? If not, did you discover anything new?